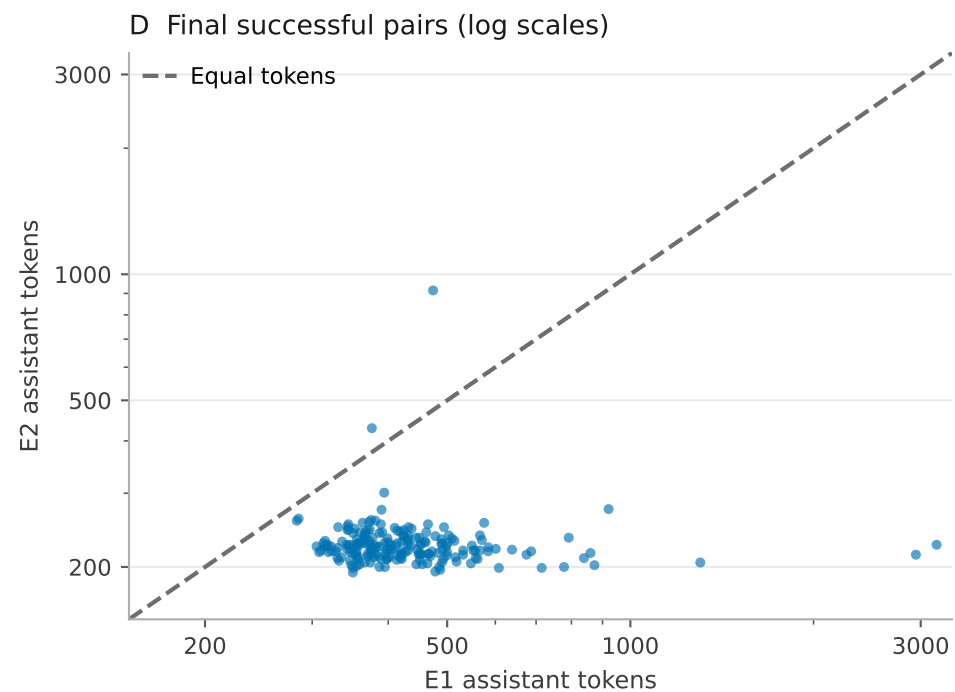
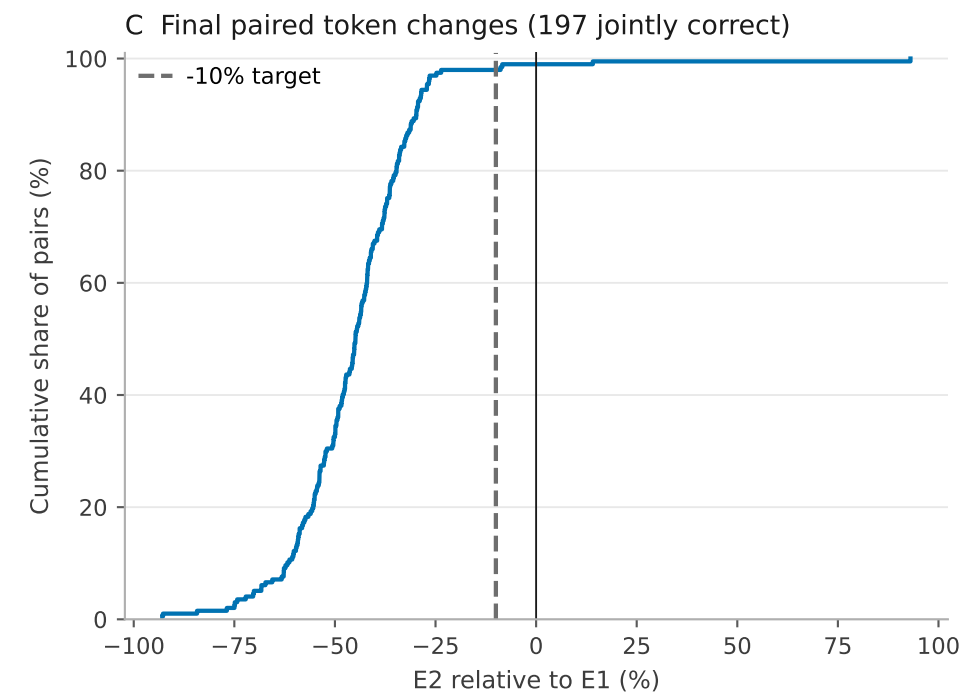
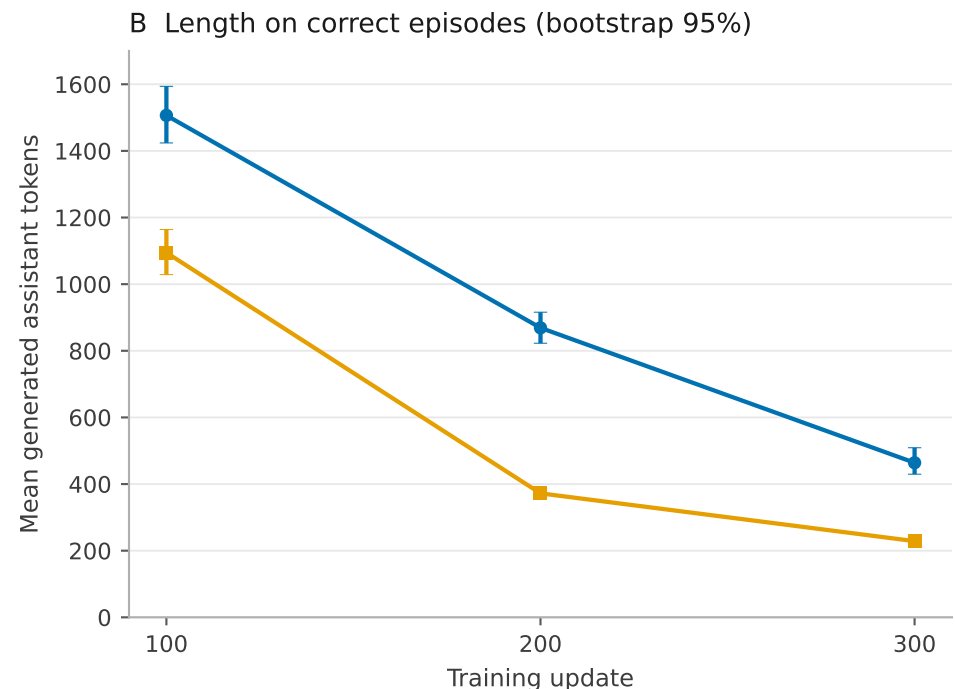
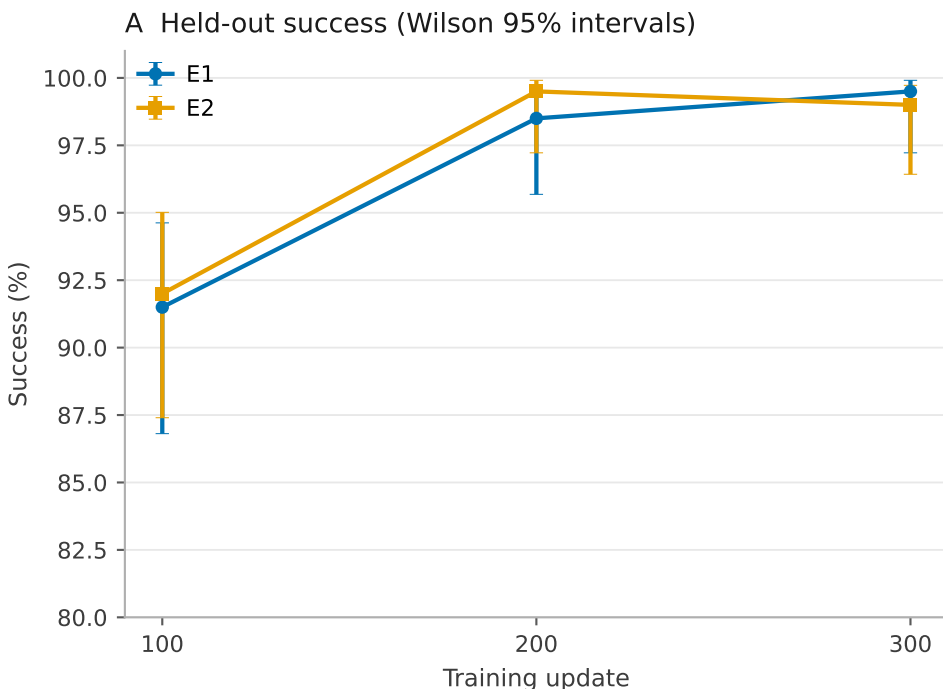
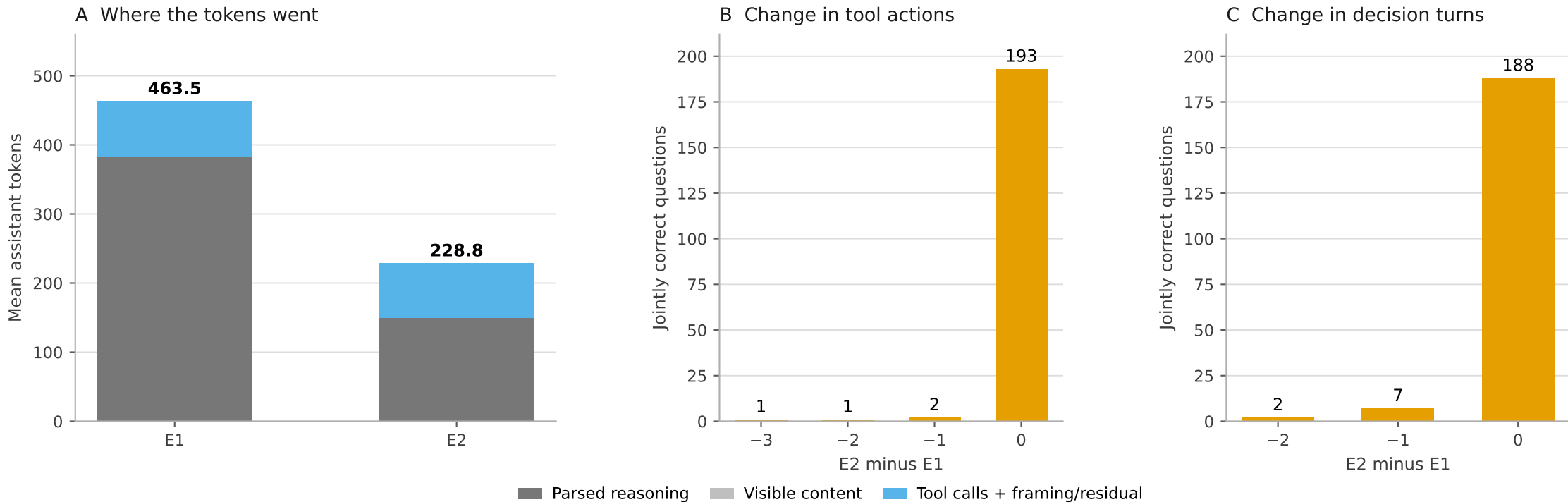


Seed 4016: one task family, one trained model per arm



Token savings come mostly from generated reasoning

Final checkpoints | Same 197 questions solved correctly by both policies | read-table-2, seed 4016



Final paired median token reduction: 44.93% (95% bootstrap interval: 42.75%-47.33%).

Parsed reasoning accounts for 99.07% of the mean saving. It measures emitted text, not internal computation.

The residual includes tool-call JSON and template framing. E2 is shorter on 195/197 jointly successful questions.

The values vary; the form and action IDs stay fixed

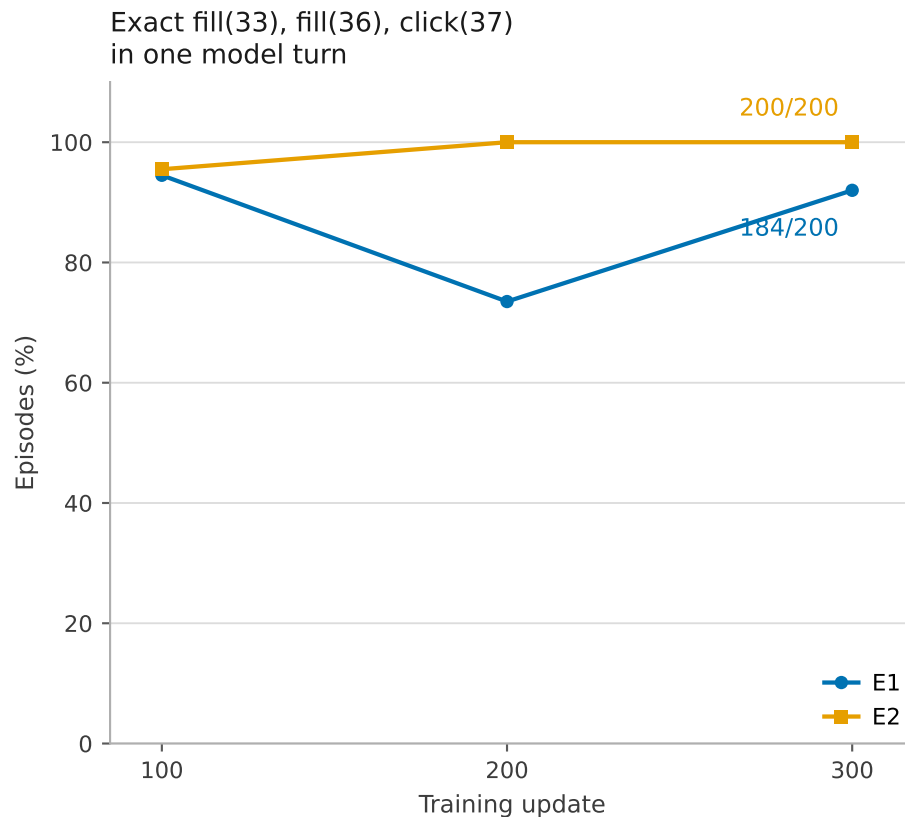
All 200 held-out pages are distinct | The same two-field lookup workflow repeats

What varies

- 200 distinct initial pages
- 56 ordered target-field pairs
- 198 table field orders
- 225 distinct target label/value pairs

What stays fixed

- Textbox IDs: 33 and 36
- Submit ID: 37
- Two values to copy, then submit



Counts include correct and incorrect episodes. At final evaluation E2 always batches two fills and Submit.
The frozen prompt asks for one tool per turn, but both arms' runtime permits batches. This shared mismatch limits interpretation.
These results do not establish adaptation to changed controls or decisions that require new information from tool feedback.

A shared action sequence, with less generated text

Final checkpoint | Episode 4: copy Religion = Judaism and First name = Meagan, then submit

E1: 378 tokens, 1 turn, 3 actions

Turn 1:
fill(33, 'Judaism')
fill(36, 'Meagan')
click(37)
Result: success

E2: 211 tokens, 1 turn, 3 actions

Turn 1:
fill(33, 'Judaism')
fill(36, 'Meagan')
click(37)
Result: success

Both final E2 failures use valid controls and submit the wrong value

Episode	Required value	E2 writes	E1 / E2 tokens
151	Arlyn	Arlin	397 / 237
182	Judaism	Punjabi	627 / 227

Episode 182 copies the Language value into Religion. E2 also fixes E1's invalid target in episode 123.

Final totals: E1 199/200, E2 198/200. Two regressions, one gain; regression upper bound 3.114% (< 5%).

Success example: nearest the median reduction among 186 pairs with one turn and three actions in both arms.
Both final E2 regressions are included. Selection is descriptive, not random. Both policies can batch calls before fill feedback.
Passing the declared 5-point success-preservation margin does not mean zero harm.

Training dynamics: trailing 20-update averages; descriptive

